Why the IEEE-14 converters are not all switched to grid-forming

The short-circuit-ratio screen and the constraint stack behind the count

In-house power system analysis toolkit

Abstract

The IEEE-14 study network carries one synchronous machine at bus 1 and four dual-mode converters at buses 2, 3, 6 and 8. When the machine trips, the supervisor promotes converters to grid-forming (GFM) one or two at a time, and never pins all four for the islanded window. This report answers why, and it leads with the short-circuit ratio.

Three results carry the argument. First, **the SCR of this island is below the WECC strong-grid floor of** 3 **on** 98.61% **of all converter-samples, and on** 100% **of the islanded window**, whose mean is 2.3015 — yet *promoting a converter cannot raise it*, because the Thevenin impedance in the definition is built from branch data, bus shunts and reference grounding alone and contains no term for the forming count. SCR is a constraint on the *plant*, not a lever the supervisor can pull. Second, **the count is bounded from both sides by constraints the supervisor cannot trade**: with no former the island is refused outright at $t = 20.0000$ s with `noVoltageFormingSource`, and with all four pinned it dies at 25.4852 s — 5.49 s after the trip, before the load step and before any fault. Third, **the intervening sets are not all available**: of the fifteen non-empty forming sets, only seven are admissible, *every* three-converter set has no equilibrium at all, and the all-four set — although admissible — is the least damped of the seven ($\zeta_{\text{worst}} = 0.1223$ against 0.2549–0.2676, less than half). The conclusion this report defends is not "fewer formers are better": it is that **the forming count is not a free parameter, stability is not monotone in it, and the SCR screen explains why adding converters does not buy the strength that is missing**.

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1 The question, and the shape of the answer

1.1 What is being asked

Four converters sit at buses 2, 3, 6 and 8, each able to run grid-following (GFL) or grid-forming (GFM). One machine at bus 1 owns the angle reference. When it trips at $t = 20\,\mathrm{s}$, at least one converter must own the reference or the island has no defined voltage angle. The natural engineering reflex is to make *all four* form the grid: maximum voltage support, maximum inertia, no follower left depending on a reference that no longer exists.

The delivered supervisor does not do that. Its forming set across the islanded window takes the values

$$1 \;\to\; 2 \;\to\; 1 \;\to\; 2 \;\to\; 4 \;\to\; 1 \;\to\; 2 \;\to\; 1, \tag{1}$$

with durations 36.98 s, 9.37 s and 92.90 s at one, two and four formers respectively, and it never performs a one-step jump from one former straight to four. **[measured]** `report_ieee14_switching_decision_basis_en`, §7.

The question is therefore precise: *why is the count chosen, rather than maximised, and what is actually wrong with switching all four?*

1.2 The answer in one paragraph

Because the quantity that is actually deficient on this island — short-circuit strength — is a property of the *network*, and the forming count is not a term in it. Adding formers cannot repair a weak grid; it changes a different question, from "can the followers lock to a stiff bus?" to "can four voltage sources share current on a network whose impedance none of them can change?". This network answers that second question badly. Measured: all four three-converter sharings have no equilibrium at all, and the four-converter sharing is admissible but the least damped of the seven admissible sets. So the count is set by the constraint stack — equilibrium existence, device limits, full-KCL small-signal margin, reference ownership and the transition certificate — and those constraints are non-tradeable, meaning a comfortable margin on one does not buy a pass on another.

1.3 Roadmap

Section 2 treats the SCR screen in detail: the definition, the source line that forces its applicability policy, the delivered measurements, and four consequences that follow from it. Section 3 lays out the constraint stack and shows, gate by gate, what refuses *all four*. Section 4 measures what pinning all four actually costs. Section 6 states the limits of the claim.

2 The short-circuit-ratio screen

2.1 Definition, exactly as implemented

For each online converter the project computes (`+stability/ibr_scr_metrics.m`, header at `:29--49`)

$$\mathrm{SCR}_i \;=\; \frac{S_{sc,i}}{S_{\mathrm{rated},i}}, \qquad S_{sc,i} \;=\; \frac{|V_i|^2}{|Z_{th,ii}|}\,S_{\mathrm{base}}, \qquad Z_{th,ii} \;=\; \left[\,\mathbf{Y}^{-1}\mathbf{e}_i\,\right]_i, \tag{2}$$

solved as `Ybus * Vz = e_k` and read at `Vz(k)` — the source comment states explicitly that this is "*never* `inv`/`pinv`" **[source]**. The matrix $\mathbf{Y}$ is built from branch data including tap and phase shift, plus bus shunts, and from nothing else:

```
No load admittance - pure network strength (PROJECT_DERIVED choice).
(ibr_scr_metrics.m:26)
```

Reference buses are grounded so that the Thevenin equivalent is taken with the voltage sources short-circuited. The rating is the converter's own `ratings.Mbase`; if it is missing or non-positive the evaluation *fails closed* rather than inferring one, and the header records that `ratings.Sbase` and `limits.Pmax_MW` are deliberately *not* accepted as fallbacks **[source]**.

The threshold and its polarity are fixed:

$$\text{pass} \;=\; \text{SCR} > 3, \qquad \text{refuse} \;=\; \text{SCR} \le 3, \tag{3}$$

classified `CASE_DEFINED` and described in the source as the "*WECC REGC_A strong-grid contract*" **[source]** — that is, the screen a transmission planner applies before allowing a *grid-following* plant to connect.

2.2 The property that decides the whole argument

Equation (2) contains no term for control mode. $\mathbf{Y}$ is branch data, bus shunts and reference grounding; $S_{\text{rated},i}$ is a nameplate number; $|V_i|$ is a measurement. Therefore:

> **Promoting a converter from GFL to GFM cannot change Z_{th}, and cannot change S_{rated}. The forming count does not enter (2) at all.**

This is not a numerical accident of one run; it is a property of the equation **[source]**. It is the single most important fact in this report, because it means the intuition "the grid is weak, therefore form more converters" cannot be executed through SCR. SCR can move only through the measured numerator $|V_i|^2$ — that is, only as a *consequence* of the island's behaviour, never as a *lever* the supervisor pulls.

Remark 1 (Scope that must travel with the number). In `ibr_scr_metrics.m:301--307` the screening profile is forced to `'not_applicable_full_state_source_model'` for the project's full-state dual family (`model_id = 'eecon49_dual'`), and the source comment explains why: this family "*models the plant explicitly, so the WECC strong-grid SCR screening profile does not apply to it*". The consequence is `eligible_for_scr = false` with `pass = true` and the recorded reason "*SCR threshold not applicable to this full-state model; equilibrium, limits and full-KCL SSSA remain mandatory*". On the delivered case all four converters report `eligible_for_scr = 0`, `SCR = NaN`, `pass = 1`. **[measured]**

The SCR figures quoted below are therefore a **published per-sample diagnostic, not an active gate**. That does not weaken the argument: the network-invariance of (2) is a property of the equation, and holds whether or not the gate is armed. It does mean the number must not be presented as a pass/fail verdict of this run.

2.3 The delivered measurements

Window	n	min	mean	max	below 3
pre-trip, machine online	168	2.0892	2.3439	2.4803	100.00%
post-trip, islanded	2760	0.6501	**2.3015**	2.8147	**100.00**%
after the load step	3152	1.5411	2.6899	3.2908	95.53%
fault at bus 9, 85.00–85.15	164	0.2944	1.2954	6.6726	97.56%
line 6–13 out	2160	2.0718	2.7319	6.6184	97.96%
restore to reclose	4384	1.8361	2.5914	3.5322	99.64%
after reclose	1928	1.9104	2.3291	2.4833	100.00%
whole run, all four converters	14716	0.2944	2.5271	6.6726	**98.61**%

Table 1: Per-sample SCR at the converter buses. Reproduced from `report_ieee14_switching_decision_basis_en`, Table 7, which reads it from the run's own `agsi_reference.scr` diagnostic; classification `ASSUMED_DIAGNOSTIC`. The ratio uses each converter's rated MVA on the system base (100 MVA), and the numerator uses the measured bus voltage of that sample. **[measured]**

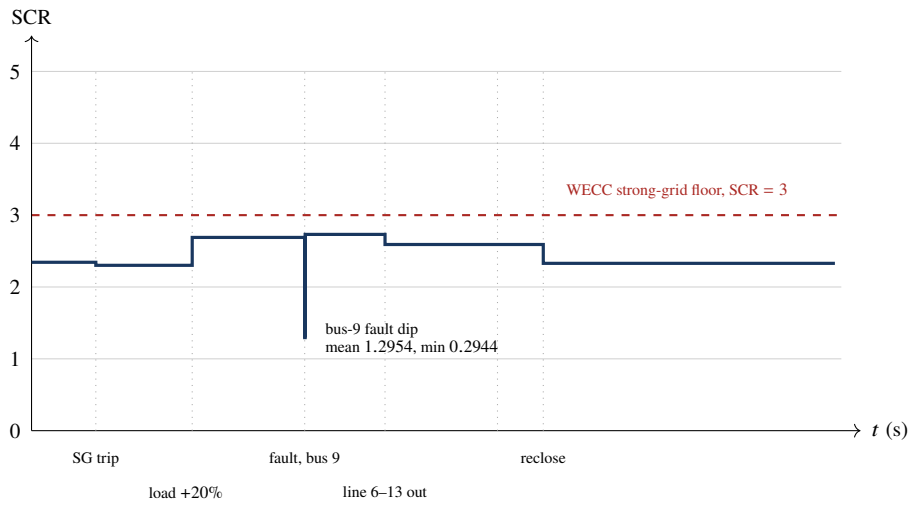


Figure 1: The mean SCR across the four converters, drawn as a step plot of the per-window means of Table 1. The dashed line is the WECC strong-grid floor. **The island never leaves the weak-grid region, in any window, at any time.**

2.4 Four consequences

1. **The machine trip barely moves SCR — and that is the point, not an error.** The window mean goes 2.3439 → 2.3015. Z_{th} is built from the passive network, and the machine is not a term in it. What changes at the trip is the numerator and, decisively, *which sources are behind the bus*: before the trip the converters are followers on a network held up by a 615 MVA machine; after it they are the only sources on the same network. **The impedance seen is the same; the support behind it is not.** That is precisely the condition grid-forming control exists to address — and precisely why the answer is "at least one must form" rather than "four must form".

2. **The island is weak by the WECC definition essentially always.** 98.61% of the delivered run's converter-samples sit at or below the floor that REGC_A uses to *refuse* a grid-following plant. The post-trip window and the post-reclose window are each 100% below it, and no window's mean reaches it. The deepest value, 0.2944, occurs during the fault, where a collapsed bus voltage is the whole of the numerator. **[measured]**

3. **Because the count cannot move** (2)**, pinning all four does not fix the weak grid.** It leaves the weakness in place and adds three more voltage sources that must now share current against the same unchanged, high network impedance. The problem changes *kind*: the REGC_A screen asks whether a PLL can lock to a stiff bus, and with all four forming there is no PLL left to protect. The screening question becomes vacuous, and is replaced by a different and harder one — multi-source current sharing on a weak network — which Section 3 shows this network answers badly. **[derived]** from (2) and **[source]**.

4. **The load step *raises* SCR, which warns against cross-window comparison.** The window mean rises 2.3015 → 2.6899 after the +20% step: load admittance is excluded from $\mathbf{Y}$, but load current raises bus voltage, and $|V|^2$ is the numerator. As implemented, (2) is a *voltage-referenced* quantity, not a pure network invariant. Comparisons are valid *within* a row of Table 1, not across rows.

Remark 2 (A rating-convention caveat, so the number is not over-read). A second diagnostic in the repository (`build_ieee14_switch_system.m:159`) computes the same $|V|^2/|Z_{th}|$ but divides by $\max(|S_{\text{bus}}|, 0.20)$ rather than by the converter rating, and reports 4.78–7.19 for the same four buses with the machine online. **Same impedance, different denominator.** The numbers in Table 1 are the ones the running supervisor's own diagnostic produces, on the rated-MVA convention, and the conclusions drawn above are conclusions about the *impedance* and the *count*, both of which are convention-independent.

3 The constraint stack behind the count

The supervisor is three stages in series and **each stage can only refuse** — no stage may overrule another's refusal, and no gate is traded against another inside a scalar. The project states this as a standing rule:

> *"Short-circuit strength, device current limits, and reference ownership are separate non-tradeable gates, never traded off against each other inside one scalar."*
>
> (`README.md:84--86`)

That sentence is the reason the answer to "why not all four?" is a *stack* of independent refusals rather than a single score.

3.1 Constraint 1 — a voltage-forming source must exist

With all four locked in grid-following, the machine is the only voltage-forming source, so opening its breaker leaves an energised island with *no angle reference*. The per-island admissibility check refuses the transaction *before any Newton solve*: the delivered `locked_gfl` arm ends at $t = 20.0000$ s after 43 samples with the identifier `noVoltageFormingSource`. **[measured]**

On this resource mix, promoting **at least one** converter to grid-forming is therefore a *necessary condition* for a post-trip trajectory to exist at all — not an improvement to one.

This is the hard *lower* bound on the count, and it is measured, not argued.

3.2 Constraint 2 — an equilibrium must exist

There are $2^4 - 1 = 15$ non-empty forming sets. The project solves every one at its own operating point before the run starts.

Set	Ref	ω (rad/s)	margin	ζ_{worst}	Outcome
$\{5\}$	5	−0.155437	+0.055437	0.2676	admissible
$\{4\}$	4	−0.252025	+0.152025	0.2629	admissible
$\{3\}$	3	−0.316711	+0.216711	0.2618	admissible
$\{2\}$	2	−0.328014	+0.228014	0.2569	admissible
$\{2,4\}$	2	−0.561723	+0.461723	0.2549	admissible
$\{3,4\}$	3	**−0.568138**	**+0.468138**	0.2560	admissible (most damped)
$\{2,3\}$	—	—	—	—	no equilibrium: IBR6 violates a limit
$\{2,5\}$	—	—	—	—	no equilibrium: IBR6 violates a limit
$\{3,5\}$	—	—	—	—	no equilibrium: IBR6 violates a limit
$\{4,5\}$	—	—	—	—	no equilibrium: IBR3 violates a limit
$\{2,3,4\}$	—	—	—	—	no equilibrium, residual 3.279×10^2
$\{2,3,5\}$	—	—	—	—	no equilibrium, residual 3.605×10^2
$\{2,4,5\}$	—	—	—	—	no equilibrium, residual 3.561×10^2
$\{3,4,5\}$	—	—	—	—	no equilibrium, residual 3.346×10^2
$\{2,3,4,5\}$	2	−0.482909	+0.382909	**0.1223**	admissible, **least damped**

Table 2: The complete SG-off admissibility enumeration, read from `summary.shared.sg_off_candidates` in `ieee14_gfm_lock_compare_zeta/summary.mat` and reproduced from `report_ieee14_switching_decision_basis_en`, Table 4. Seven of fifteen sets are admissible. ω is the least damped member of the frozen finite-difference set; margin $= -\omega - \gamma_{\text{req}}$; ζ_{worst} is the worst damping ratio over the physical decision spectrum. This is a *base-load table, frozen before candidate evaluation*. **[measured]**

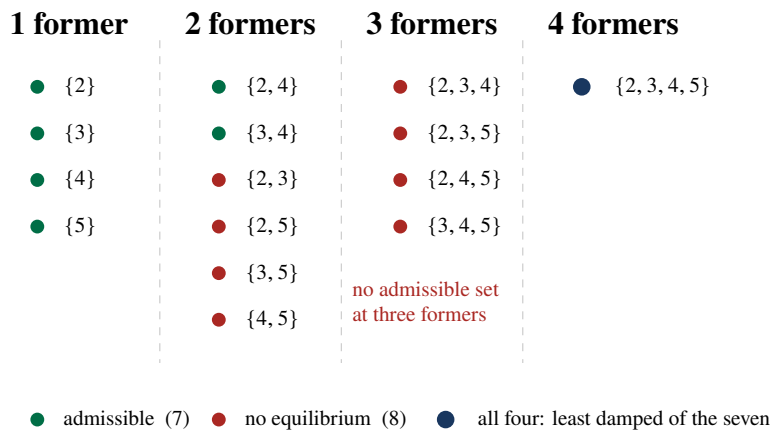


Figure 2: The admissibility map. **Feasibility is not monotone in the number of forming converters**: every admissible set sits below or above the three-unit row, never on it. Going from two units to three is not a compromise — it lands where the equations have no solution at all.

Three facts here are load-bearing.

1. **Eight of fifteen sets do not exist as operating points.** Four fail because a converter left in grid-following is asked to violate an operating limit; four fail because the four-way power balance has no solution at all (coupled-Newton residuals 3.28–3.61 $\times 10^2$). Each failure is *reported by name*, not as a low margin.
2. **All four three-converter sets fail.** The middle of the map is empty at every one of the four ways of choosing three. A policy that climbed the count one at a time would be refused at the third step.
3. **The all-four set is admissible but is the outlier.** Its worst damping ratio, 0.1223, is *less than half* of every other admissible set's 0.2549–0.2676, and it is the only admissible set whose worst ratio is not comfortably above the floor. It passes the gate — and is the worst of the seven that pass.

3.3 Constraint 3 — the damping-ratio floor, at three perturbation sizes

The acceptance criterion is declared by the case, not chosen by this report: `acceptance_criterion = 'damping_ratio_floor'`, with `zeta_min_damping` = 0.05 and `frozen_before_candidate_eval` = 1 **[source]**. A candidate is admissible only when

$$\forall \varepsilon \in \{1.5,\ 3,\ 6\} \times 10^{-6}: \quad \max \operatorname{Re} \lambda(\varepsilon) < 0 \quad \text{and} \quad \min_i \zeta_i(\varepsilon) \ \geq\ 0.05, \qquad \zeta_i = \frac{-\operatorname{Re} \lambda_i}{|\lambda_i|}. \tag{4}$$

Three features of (4) matter for the count question.

- **It is a floor on the *worst* mode, not on the rightmost.** The worst-damped mode is generally not the rightmost one, so reading the margin off $\max \operatorname{Re} \lambda$ would miss it.
- **It is a *ratio*, so it is timescale-invariant.** This plant carries modes from roughly 0.02 Hz — the island's electromechanical swing, slowed by the converters' small inertia — up to the current-loop band, a span of more than three decades. No single absolute decay rate is a meaningful floor across that span; a ratio is.

- **All three perturbation sizes must agree.** Disagreement fails closed. A margin that exists at one finite-difference step but not at the next is not a margin.

The separate quantity $\gamma_{\text{req}} = 0.1$ rad/s is the *ordering key* among already-admissible candidates only — it is not the gate. This is why a set can be admissible while its `margin` is negative, as IBR8's is at margin = +0.055437 against a gate that never consults it.

3.4 Constraint 4 — the count is chosen as a minimum, not a maximum

The runtime rule is the following, and it is the direct answer to the count question.

Proposition 1 (Minimum-cardinality admissible superset)**.** *Let $\mathcal{F}$ be the forming set at the decision instant and $\mathcal{T}$ the set of table rows with* `feasible` *and* `ready_to_commit` *both true. The returned destination is the row $c \in \mathcal{T}$ minimising, in order: (i) $|\text{selected}(c)|$; (ii) $-\text{margin}(c)$; (iii) the selected index tuple lexicographically; (iv) the reference index — subject to $\mathcal{F} \subsetneq \text{selected}(c)$, i.e. the destination must be a* strict *superset of the current set (*`select_support_augmentation_candidate.m:54--58, 101--112`*). If no such row exists the function returns* `found=false` *and the supervisor holds its incumbent set; it never fabricates a candidate and never relaxes a gate.*

Two consequences follow immediately. **The cardinality objective is a minimum**: nothing in Proposition 1 rewards adding converters, and a three-unit destination is chosen over a two-unit one *only* when no two-unit strict superset is admissible. And **the choice is not a free re-selection**: because the destination must contain the current set, the supervisor cannot answer "the island needs four formers" by promoting four at once from one; it can only add to what it has, and it adds as few as the table permits. On the delivered chronology the resolution is

t (s)	incumbent $\mathcal{F}$	candidate supersets	admissible	chosen
22.0887	$\{2\}$	all seven supersets of $\{2\}$	$\{2,4\}$, $\{2,3,4,5\}$	$\{2,4\}$
51.3403	$\{2\}$	(as above)	$\{2,4\}$, $\{2,3,4,5\}$	$\{2,4\}$
53.3795	$\{2,4\}$	$\{2,3,4\}$, $\{2,4,5\}$, $\{2,3,4,5\}$	$\{2,3,4,5\}$ only	$\{2,3,4,5\}$
148.2811	$\{2\}$	(as at 22.0887)	$\{2,4\}$, $\{2,3,4,5\}$	$\{2,4\}$

Table 3: Every augment of the delivered chronology resolved through Proposition 1 using the admissibility of Table 2. **[measured]**

At $t = 22.0887$ s and $t = 51.3403$ s two destinations exist and the smaller one wins on criterion (i). At $t = 53.3795$ s — after the +20% load step, with the island carrying the machine's 133.19 MW deficit — the two-unit supersets $\{2,3,4\}$ and $\{2,4,5\}$ are among the eight rows with no equilibrium, so the only admissible destination is the all-four set and criterion (i) has nothing to choose. **Four forming converters is not a policy preference; it is the residue of the enumeration after the load step.**

> *The four-unit promotion happens because at +20% load nothing smaller can be certified — not because four was judged better than two.*

3.5 Constraint 5 — the transition, not just the destination

Stage 3 re-simulates the would-be committed state with the *same* integration kernel before committing, over a horizon derived from the destination's own settling time; a refusal is atomic and the previous accepted state stands. Over the delivered 250 s the certificate was consulted at four transactions and changed the outcome at one — at $t = 53.3795$ s, where the untouched arrival was rejected and the conditioned variant committed at a peak excursion of 15.495°. **[measured]**

This matters for the count question because **admissibility of all four is a statement about an equilibrium, and says nothing about the trajectory that arrives there**. Section 4 measures a case in which the equilibrium was admissible and the arrival was not — a failure no eigenvalue check can see.

4 What switching all four actually costs

4.1 The six delivered arms

Arm	horizon reached	samples	termination
`locked_gfl`	20.0000 s	43	`noVoltageFormingSource`
`pinned_gfm4`	**25.4852 s**	837	`adaptiveDtMin`, $\Delta t_{\min} = 6.104 \times 10^{-6}$
`no_adaptation`	250.0000 s	1565	`SUCCESS`, reclose 151.1321 s
`pinned_gfm1`	250.0000 s	1766	`SUCCESS`, reclose 151.1321 s
`pinned_gfm2`	250.0000 s	1761	`SUCCESS`, reclose 154.0244 s
`adaptive`	250.0000 s	3679	`SUCCESS`, reclose 159.2519 s

Table 4: The six delivered arms, from `figures/switch_ieee14_decision/comparison_summary.tex`. All six share one network, one dispatch, one event schedule, one solver and one step control, and the runner asserts the arms differ only in the fields each arm declares. They are bit-identical on $[0, 20)$ s over 42 samples, so the difference after that instant is attributable to the control policy and to nothing else. `pinned_gfm4` pins the forming set to all four. **[measured]**

Both endpoints are wrong, and both are measured. With no former at all the island dies at $t = 20.0000$ s — the trip instant itself. With all four pinned it dies at 25.4852 s, 5.49 **s after the trip, before the load step and before any fault.** **[measured]**

Remark 3 (Pinning is not the defect). `pinned_gfm2` pins two and reaches 250 s with a successful reclose. The evidence condemns the *count of four*, not the pinned mode.

4.2 How all four fails

The terminal state of `pinned_gfm4` is not a gradual loss of damping. In its last accepted sample the four converters have separated in frequency:

	IBR2	IBR3	IBR6	IBR8
frequency (Hz)	**64.9947**	59.1383	59.1809	59.1946

A 5.86 Hz split, with one converter having run away from the other three. The step controller then reported 398 successive Newton non-convergences and terminated at the minimum step. Reclose was never requested. **[measured]**

4.3 Why an admissible set can still die

The acceptance gate (4) is established by linearising about the solved operating point. The defect record (`docs/project/defects/2026-08-14-all4-gfm-commit-outside-basin.md`) measured the gap by re-running the same inputs from two different initial device states:

Arm	initial device state	outcome	excursion
ARM A	the state the production run actually arrives with	**SLIPS**	4358°
ARM B	the certified equilibrium state of the same configuration	**SYNC**	6.6°

Table 5: The same committed configuration, the same inputs, two initial device states. The carrying coordinate is IBR2's outer voltage-loop integrators gfm_xi_Vd/xi_Vq: 0.144253 on arrival against 0.199658 at the certified equilibrium — a 28% displacement in a coordinate *no bus quantity reflects*. **[measured]**

The mechanism is worth stating because it is the reason "more formers" is not merely unhelpful but actively harder. The unit that carries the failure is the *incumbent* former — the one that has been grid-forming since $t = 20$ s and therefore does *not* pass through a mode transfer at the commit. The transfer contract conditions the devices that change mode and leaves the incumbent's integrators untouched, while the same transaction simultaneously moves every former to a new certified operating point. The more converters a single transaction must re-seat simultaneously, the larger that uncontrolled displacement becomes.

Remark 4 (Two failure modes, and they are different). The all-four row of Table 2 fails a test of the *trajectory*, while the three-unit rows fail the weaker test of *existence* — the equilibrium does not exist at all. A policy that asked only "is the destination stable?" would admit all four and lose the island, which is exactly what the pre-certificate baseline did at $t = 53.4025$ s. This is why Stage 3 re-simulates before committing, and why the certificate is a forward trial rather than a second eigenvalue check: no eigenvalue test can see the arriving device state.

4.4 Three claims and their scopes

The honest statement is narrower than "four is bad". Three separate statements, each separately evidenced:

1. **It is the least damped admissible configuration.** $\zeta_{\text{worst}} = 0.1223$ against 0.2549–0.2676 for the other six, a factor of about two. Measured, and a property of the *equilibrium* (Table 2).
2. **Its arrival is not robust.** Measured, and a property of the *trajectory* (Table 5). This is the failure mode that actually ended the pinned arm.
3. **Where it *is* chosen, it is chosen as a residue.** At $t = 53.3795$ s the all-four set was the only admissible destination (Table 3). The policy did not prefer four over two; it had no alternative.

Against those three, one measured statement points the other way and belongs in the same list, because a report that omits it is not a proof:

> In the defect record's load-severity experiment, the all-four configuration rides a 2.0 pu load deficit that the two-former configuration cannot — two formers slip at the same deficit. **[measured]**

So the defensible conclusion is not "fewer formers are better":

> *The number of forming converters is not a free parameter and stability is not monotone in it. Both endpoints are wrong: with no former the island dies in* 20 *s, and with four pinned the island dies in* 25.5 *s. The count has to be chosen per configuration, from the certified margin and the arrival, and the ranking has to be allowed to prefer two to four when two is certified.*

5 Summary: the answer in constraint order

#	Constraint	What it does to the count
1	Voltage-forming source exists	Forces the count ≥ 1. All-GFL is refused at t = 20.0000 s with `noVoltageFormingSource`; the island has no angle reference at all.
2	Equilibrium exists	Removes 8 of 15 sets. *Every* three-unit set has no equilibrium; four two-unit sets fail on a device limit. This is the binding constraint in this case, not the damping floor.
3	Damping-ratio floor (4)	Admits all four, but records it as the least damped of the seven (ζ_{worst} = 0.1223, less than half the others).
4	Minimum-cardinality rule	Makes the count a *minimum* subject to the above. Four is chosen only at t = 53.3795 s, where nothing smaller is admissible — as a residue, not a preference.
5	Transition certificate	Tests the *arrival*, not only the destination. The all-four configuration is admissible and its arrival is outside the basin (4358° against 6.6°).
SCR screen (constraint on the plant, not the count)		98.61% of samples below the floor of 3; the count cannot enter (2). Pinning all four therefore *cannot* repair the weak grid — it only adds voltage sources that must share current against the same unchanged impedance.

Table 6: The constraint stack, in the order the supervisor applies it. Each gate is reported by name and none is tradeable against another.

The question "why not all four?" therefore has a two-part answer. Part one is the SCR result: the deficiency that motivates adding formers is short-circuit strength, and the forming count is not a term in it, so adding formers does not address the deficiency — it converts a PLL-locking

problem into a multi-source current-sharing problem on a network measured weak in every window. Part two is the constraint stack: the count is a minimum over admissible configurations, three-unit configurations do not exist at all, and the one configuration that would switch everything is the least damped of the seven that do, is chosen only as a residue after the load step, and fails on the *arrival* even though its equilibrium passes.

6 What this report does not claim

1. **It does not claim two formers are always better than four.** The defect record contains a measured case where all four rides a 2.0 pu deficit that two formers cannot. The claim is that stability and robustness are not monotone in the count, in both directions.
2. **It does not claim the SCR values are plant-strength ratings.** Equation (2) is voltage-referenced as implemented, the gate is exempt for this model family, and a second repository diagnostic gives 4.78–7.19 for the same buses under a different rating convention. Comparisons are valid within the convention of Table 1.
3. **It does not claim the SCR screen made the switch decision.** For `eecon49_dual` the screen is *not applicable* and does not gate. SCR enters this report as the *physical* explanation of why adding formers does not buy strength — not as the mechanism that refused a transaction.
4. **It does not claim the admissibility table is load-independent.** Table 2 is the frozen base-load table. The +20% load findings quoted in Section 4 are a separate measurement from the defect record, at a different load level.
5. **It does not claim the basin statement is general.** The 4358° against 6.6° contrast is measured on one transition pair; the mechanism is offered as an explanation of that pair, not as a law. The defect record states plainly that it does not root-cause the `pinned_gfm4` death.
6. **It does not re-run anything.** The numerical caches of the delivered result set are build outputs and are not present in this worktree. Every figure here is attributed to the delivered artefact that already carries it, so it can be re-derived from a regenerated run.

7 Reproduction

All commands are run from the repository root.

```
% 1. the six-arm comparison that produces the shared admissibility table
matlab -batch "run('scripts/reporting/run_ieee14_gfm_lock_comparison.m')"

% 2. this report
cd docs/source
xelatex report_ieee14_all_gfm_constraints_en.tex
xelatex report_ieee14_all_gfm_constraints_en.tex
```

The regression gate for the delivered adaptive arm is the seven-value row of the companion report: horizon 250.000000 s, 3679 samples, 161 rejected steps, maximum accepted KCL residual 9.972081×10^{-9}, terminal f_{COI} 60.000001128 Hz, reclose time 159.2519 s with status `SUCCESS`, and handback completion at 173.1271 s. If a re-run does not reproduce all seven, the numbers in this report do not apply to it.